

EXPERIENCE

Senior Machine Learning Engineer II at Thomson Reuters

Jul 2021 - Present

Toronto, Canada

Senior Research Engineer at TR Labs, Thomson Reuters' applied research division, focused on CLEAR — the company's flagship due diligence platform for KYC/AML screening and investigations. Built production ML systems across the full lifecycle — including a commercially launched agentic investigation product, LLM-powered summarization features, and large-scale data infrastructure — in a regulated environment where model accuracy has direct compliance consequences.

NLP & Machine Learning Engineer at INAGO INC.

Jul 2018 - Jun 2021

Toronto, Canada

Built NLP models for language understanding and automated text generation, including fine-tuning BERT and T5 for domain-specific tasks. Managed the full training lifecycle and led collaborative research projects with university partners.

EDUCATION

M.Sc. in Computer Science, York University

2014 - 2018

Toronto, Canada · GPA: 8.17 / 9

Thesis: Interactive Question Answering Using Frame-based Knowledge Representation

B.Sc. in Computer Engineering, Amirkabir University of Technology

2010 - 2014

Tehran, Iran · GPA: 17.18 / 20

PROJECTS

Agentic Investigation System — CLEAR Investigate

2025 - 2026

Contributed to building CLEAR Investigate, Thomson Reuters' first agentic AI product, live in production for customers. Designed the agent experimentation architecture using PydanticAI for rapid prototyping of multi-step workflows across entity search, report retrieval, web search, and an internal knowledge graph. Implemented LLM caching (~25% cost reduction), automated LLM-as-judge evaluation pipelines, and SME annotation workflows for ground-truth benchmarking.

Python · PydanticAI · OpenAI GPT-4 · Claude Sonnet · LLM-as-Judge Evaluation · AWS

Entity Resolution Data Infrastructure — CLEAR KYC/AML Platform

2022 - 2024

Designed and built the data engineering infrastructure for an entity resolution system operating across ~800M entities and billions of documents at 700 idents/second. Architected a PII-isolated AWS environment and a unified versioned schema reconciling two incompatible annotation sources spanning hundreds of thousands of labeled records. Built Spark and Python pipelines for multi-format merging, conflict signal surfacing, and model experimentation, later automated into SageMaker Pipelines.

Spark · Python · Scikit-Learn · AWS SageMaker · CloudFormation · S3 · EMR · OpenSearch · RDS

Automated Question Generation from Documents

2020

Fine-tuned T5 transformer for automated question generation, cutting manual data curation effort by 40%. Experimented with model input representations and evaluated using BLEURT as part of a university collaborative research project.

PyTorch · SpaCy · HuggingFace Transformers · Python

Conversational Question Answering

2018

Built a domain-specific question answering dialogue system using syntactic and semantic document analysis and ontology generation, as part of a collaborative industry research project.

SpaCy · Python

CLEAR Business AI — GenAI Report Summarization

2024

Built the AI-powered summary panel for CLEAR Business entity reports, live in production processing hundreds of reports daily. Designed a selective XML parsing engine and multi-prompt LLM architecture with separate calls for business overview, risk analysis, and social media discovery — with Bing Search integration and a verification LLM call to filter false positives. Implemented citation linking to source locations and ran SME annotation and evaluation rounds before launch.

Python · OpenAI GPT-4 · Claude Sonnet · Bing Search API · Prompt Engineering · XML Processing

Semantic Search Improvement — Checkpoint Tax Research

2021

Shipped a semantic search improvement for Checkpoint, Thomson Reuters' tax research platform, improving access to conceptual documents by 95%. Built a query intent classifier using sentence embeddings to dynamically promote higher-level content for general queries, delivered to production within a ~100ms latency budget.

Python · Sentence Embeddings · NLTK · Elasticsearch · AWS · Java

Domain-Specific Language Understanding Engine

2019

Trained domain-specific Word2Vec embeddings and LSTM-based NLU models with interpretability testing to improve language understanding accuracy and model transparency.

PyTorch · Python · Word2Vec · LSTM

## | PUBLICATIONS

---

### Question-worthy Sentence Selection for Question Generation

Canadian AI 2020, 2020 · co-authored

### Interactive Question Answering Using Frame-based Knowledge Representation

York University M.Sc. Thesis, 2018

### Time Aware Topic-based Recommender System

Big Data & Information Analytics, 2016 · co-authored

### A Study on Prediction of User's Tendency Toward Purchases in Websites based on Behavior Models

Information and Knowledge Technology (IKT), 6th Conference, IEEE, 2014 · co-authored

## | TECHNICAL SKILLS

---

Python · PyTorch · Scikit-Learn · Numpy · Pandas · SpaCy · NLTK · Transformers · Large Language Models · Prompt Engineering · Agentic AI · PydanticAI · Retrieval-Augmented Generation (RAG) · LLM Evaluation · NLP · Machine Learning · Deep Learning · XGBoost · Apache Spark · AWS · Amazon SageMaker · AWS CloudFormation · Docker · PostgreSQL · Elasticsearch · Opensearch · OpenAI API · Java · Scala · Bash · Agile Development